

REVIEW

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Green synthesis of silver and zinc oxide nanoparticles using *Ziziphus jujuba*: mechanisms, characterization, applications, and future perspectives

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Abstract

The escalating demand for sustainable, non-toxic nanomaterials has driven interest in plant-mediated green synthesis routes that avoid hazardous chemical reagents. *Ziziphus jujuba* Mill. (Jujube, Rhamnaceae) presents an underexplored system despite its phytochemical diversity flavonoids (quercetin, rutin, and luteolin), phenolic acids (chlorogenic acid, caffeic acid), triterpenoids (betulinic acid, oleanolic acid), vitamin C, alpha-tocopherol, and polysaccharides that collectively satisfy the mechanistic requirements for nanoparticle synthesis. This review applies a structured literature search strategy (Scopus, Web of Science, and PubMed; 2015–2025) to critically examine *Z. jujuba*-mediated synthesis of silver nanoparticles (AgNPs) via phyto-reduction and zinc oxide nanoparticles (ZnO NPs) via bio-mediated co-precipitation and thermal decomposition. The review covers: (i) the phytochemical basis and mechanistic pathways of synthesis; (ii) the influence of synthesis parameters precursor concentration, extract ratio, pH, temperature, reaction time, and calcination temperature on nanoparticle properties; (iii) characterization by UV-Vis, XRD, FTIR, SEM/TEM, DLS, and zeta potential; and (iv) applications including antimicrobial activity against multidrug-resistant (MDR) pathogens, photocatalytic dye degradation, antioxidant activity, anticancer potential, and biosensing. A critical evaluation across studies reveals substantial variability in reported particle size (8–90 nm for ZnO NPs depending on synthesis route) that has not been systematically attributed to calcination temperature, extract ratio, or phytochemical batch variation. The review identifies that no published study has performed a controlled, same-batch synthesis of both AgNPs and ZnO NPs from *Z. jujuba* extract with direct mechanistic and functional comparison. This gap, together with the limited calcination-temperature optimization, scale-up data, ecotoxicological assessment, and in vitro cytotoxicity profiling, defines priority directions for future research in this field.

Keywords *Ziziphus jujuba*, Silver nanoparticles, Zinc oxide nanoparticles, Green synthesis, Antimicrobial, Photocatalysis

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1 Introduction

The convergence of nanotechnology and green chemistry has given rise to one of the most productive and rapidly expanding research areas of recent decades. Metal and metal oxide nanoparticles, particularly silver nanoparticles (AgNPs) and zinc oxide nanoparticles (ZnO NPs), have become key materials in this convergence owing to their physicochemical properties and broad application profiles [1, 2]. AgNPs exhibit a strong and tunable surface plasmon resonance (SPR) optical response, broad-spectrum antimicrobial activity, and utility in colorimetric and surface-enhanced Raman scattering (SERS) sensing platforms [3, 4]. ZnO NPs are wide-band-gap semiconductors (3.37 eV) with a high exciton binding energy (60 meV) and have been applied in UV-driven photocatalysis, antimicrobial activity, electrochemical sensing, and reactive oxygen species (ROS)-mediated anticancer applications [5, 6]. In the context of antimicrobial resistance (AMR) and industrial wastewater contamination, both nanoparticle systems are increasingly explored as alternatives to conventional antibiotics and chemical oxidants [7, 8].

Conventional chemical synthesis routes for these nanomaterials rely on hazardous reagents sodium borohydride, hydrazine, and cetyltrimethylammonium bromide (CTAB) for AgNPs, and strong bases combined with high-temperature calcination for ZnO NPs raising environmental, occupational health, and regulatory concerns [9, 10]. Green synthesis, which uses the phytochemical content of plant extracts as simultaneous reducing, capping, and stabilizing agents, offers an alternative to these routes [11, 12]. Following the articulation of the principles of green chemistry by Anastas and Warner [13], plant-mediated nanoparticle synthesis has developed from a proof-of-concept approach into an established research area [14, 15]. Plant extracts containing phenolics, flavonoids, terpenoids, ascorbic acid, and polysaccharides can perform the three roles required for nanoparticle formation ionic reduction, nucleation control, and colloidal stabilization without additional reagents [16, 17].

Ziziphus jujuba Mill. (Rhamnaceae) is a deciduous tree native to temperate and subtropical Asia with a recorded medicinal history of over 3,000 years [18, 19], commonly known as jujube, Chinese date, or red date. The leaves of *Z. jujuba* have a structurally diverse phytochemical profile: using ultra-high-performance liquid chromatography (UHPLC), Xue et al. [20] detected 13 phenolic acids and flavonoids across different *Z. jujuba* tissue types, with quercetin and rutin the most abundant in leaves, and (+)-catechin, chlorogenic acid, gallic acid, and luteolin also consistently detected. Reports on *Z. jujuba* leaf extract composition also describe alphatocopherol, beta-carotene, ascorbic acid, sterols, tannins,

alkaloids, saponins, and polysaccharides [21]. Together, these compound classes provide reducing, chelating, and capping functions relevant to AgNP and ZnO NP formation, although the precise proportions vary with cultivar, growing region, and extraction conditions (see Sect. 2.3) [22, 23].

Compared with extensively studied plants such as *Azadirachta indica*, *Curcuma longa*, and *Ocimum sanctum*, for which hundreds of nanoparticle synthesis studies exist [3, 4, 7], *Z. jujuba* remains comparatively underexplored. Fewer than fifteen primary studies have reported the use of *Z. jujuba* extract for nanoparticle synthesis [22–27]. Among these, no study has synthesized both AgNPs and ZnO NPs from the same batch of *Z. jujuba* extract under controlled, systematically optimized conditions [22–27]. This represents a methodological gap that limits direct mechanistic and functional comparison between the two nanoparticle systems.

This review aims to address this gap by (i) systematically analyzing the phytochemical basis of *Z. jujuba*-mediated synthesis of AgNPs and ZnO NPs; (ii) critically comparing synthesis parameters and their effects on nanoparticle properties; (iii) evaluating characterization data; (iv) surveying applications with quantitative performance data, distinguishing direct *Z. jujuba* evidence from evidence drawn from related plant systems; and (v) identifying literature gaps and corresponding research directions. To our knowledge, this is among the first reviews to integrate *Z. jujuba*-AgNPs and *Z. jujuba*-ZnO NPs within a single critical analysis.

1.1 Literature search strategy

We conducted systematic searches of three databases (Scopus, Web of Science [Core Collection], and PubMed) and used Google Scholar to identify grey literature and recent preprints. The search covered the period from January 2015 to October 2025, with particular emphasis on publications from 2022 onwards. The following Boolean search strings were applied to titles, abstracts, and keywords: (1) (“*Ziziphus jujuba*” OR “*Z. jujuba*” OR jujube) AND (“green synthesis” OR “biosynthesis” OR “photosynthesis”) AND (“silver nanoparticles” OR “AgNPs” OR “zinc oxide nanoparticles” OR “ZnO NPs” OR “ZnO nanoparticles”); (2) (“*Ziziphus jujuba*” OR jujube) AND (nanoparticle OR nanomaterial) AND (antimicrobial OR antibacterial OR photocatalytic OR anticancer OR antioxidant OR biosensor); (3) (“plant extract” OR “leaf extract”) AND (“AgNPs” OR “ZnO NPs”) AND (“calcination temperature” OR “characterization” OR “Scherrer”).

Inclusion criteria were: (i) peer-reviewed original research articles, reviews, or short communications; (ii) studies reporting green/biological synthesis of AgNPs and/or ZnO NPs using plant extracts, preferably *Ziziphus* species; (iii) studies reporting at least one

characterization technique (UV-Vis, XRD, FTIR, SEM/TEM, DLS, or BET); (iv) publications in English from 2015 to 2025. Exclusion criteria were: (i) conference abstracts without full text; (ii) studies on chemically or physically synthesized nanoparticles without a green-synthesis component; (iii) duplicate reports of the same dataset; (iv) sources from journals not indexed in Scopus, Web of Science, or PubMed at the time of search, applied as a proxy quality filter.

The initial search returned approximately 1,840 records across the three databases. After removing duplicates (612), title/abstract screening excluded 1,062 records that did not meet the inclusion criteria, leaving 166 records for full-text assessment. Of these, 103 met all inclusion

Table 1 Phytochemical classes of *Ziziphus jujuba* leaf extract and their proposed mechanistic roles in AgNP and ZnO NP green synthesis

Compound Class	Key Compounds	Reducing Role	Capping/Stabilizing Role
Flavonoids	Quercetin, Rutin, Luteolin, Catechin	Primary. Oxidation of phenolic –OH to quinone donates electrons for $\text{Ag}^+ \rightarrow \text{Ag}^0$ and Zn^{2+} chelation	Quinone surface adsorption; limits agglomeration
Phenolic Acids	Chlorogenic acid, Caffeic acid, Gallic acid	Secondary. Carboxylate and –OH groups donate electrons; enol tautomerism releases H^+	Carboxylate groups coordinate to NP surface; electrostatic repulsion
Triterpenoids	Betulinic acid, Oleanolic acid	Minimal (few reducing groups)	Alkyl chains may form a steric barrier limiting inter-particle contact
Polysaccharides	Jujube polysaccharides (ZJP)	Minor (hydroxyl groups)	Macromolecular corona; may contribute to electrosteric stabilization
Vitamins	Ascorbic acid, α -Tocopherol	Ascorbic acid readily oxidised to dehydro-ascorbic acid; effective electron donor	Tocopherol may anchor to surface via phenolic group
Tannins	Hydrolysable tannins	Secondary reducing capacity	Polyphenolic –OH groups may coat NP surface, reducing oxidation
Alkaloids	Julifloridine, Nummularine	Minimal	Nitrogen lone pairs may weakly coordinate Zn^{2+} ; minor role proposed

criteria and form the evidence base for this review (Figure A1, PRISMA-style flow diagram). Within this set, twelve records specifically used *Z. jujuba* extract for AgNP and/or ZnO NP synthesis, consistent with the ‘fewer than fifteen primary studies’ statement in Sect. 1. This systematic process substantiates the novelty and gap statements made throughout the review, while the gap statements themselves have been reworded in qualified form (Sect. 8).

2 *Ziziphus jujuba*: botanical profile and phytochemical composition

2.1 Botanical description

Ziziphus jujuba Mill. (Syn. *Ziziphus zizyphus*) is a deciduous tree or shrub of the family Rhamnaceae that grows to 5–12 m in height [18]. It is native to temperate and subtropical regions of Asia, including China, India, and the Middle East, and is now cultivated across Africa, Australia, and the Mediterranean [19]. The plant is adapted to dry and semi-dry environments and has been used in traditional medicine across its distribution range as a tonic, sedative, wound-healing agent, and hepatoprotective agent [28]. These traditional uses have been examined in modern phytochemical studies that have characterized its secondary metabolite profile [20, 21].

2.2 Phytochemicals and nanoparticle-forming capacity

The phytochemical profile of *Z. jujuba* leaves is notable both for its diversity and for the relevance of individual compound classes to nanoparticle synthesis. Xue et al. [20] characterized 20 cultivars and three plant tissues for total phenolic and flavonoid content, and found that leaves contained the highest total phenolic and flavonoid content of all tissues examined a finding relevant to green synthesis, where leaves are the principal extraction material. Seven of the compounds identified were detected in every tissue and cultivar examined [20]. Guo et al. [29] used UPLC-DAD-MS to confirm the presence of nucleosides and nucleobases in *Z. jujuba* fruits, and reports on *Z. jujuba* leaf extract describe the presence of alpha-tocopherol, beta-carotene, phenolic compounds, flavonoids, sterols, tannins, alkaloids, saponins, and fatty acids [21], representing a range of compound classes with reducing, stabilizing, and capping functions relevant to nanoparticle synthesis. Table 1 presents the phytochemical classes of *Z. jujuba* leaf extract and their proposed mechanistic roles in the green synthesis of AgNPs and ZnO NPs.

The *Z. jujuba* extract system is characterized by the simultaneous presence of hydrophilic reducing agents (ascorbic acid, chlorogenic acid, gallic acid) and lipophilic capping agents (alpha-tocopherol, betulinic acid) [21]. This combination has been proposed to support rapid reduction of Ag^+ and complexation of Zn^{2+} by the

hydrophilic fraction, while the lipophilic fraction may passivate the growing nanoparticle surface through hydrophobic interactions [22, 23]. While dual hydrophilic–lipophilic phytochemical profiles are not unique to *Z. jujuba* and have been reported in other plant extract systems, the specific combination present in *Z. jujuba* leaves has received limited direct experimental investigation for nanoparticle synthesis [21, 24].

2.3 Phytochemical variability and implications for synthesis reproducibility

The composition of *Z. jujuba* leaf extract is not fixed and is influenced by several factors that directly affect its performance as a reducing and capping agent in nanoparticle synthesis.

Cultivar variation Xue et al. [20] found that seven phenolic acids and flavonoids were common to all 20 cultivars studied, but the relative proportions of quercetin, rutin, and chlorogenic acid differed considerably between cultivars, and total flavonoid content varied by more than two-fold across the cultivar panel.

Geographical origin and growing conditions Phenolic and flavonoid accumulation in *Ziziphus* leaves is known to respond to soil composition, water availability, and altitude factors that vary considerably between the production regions (e.g., China, Saudi Arabia, North Africa) from which extracts used in the reviewed studies were sourced [20, 21].

Harvesting season Phenolic content in deciduous *Ziziphus* leaves is generally highest during active vegetative growth and declines as leaves senesce. As a result, extracts prepared from leaves collected at different times of year may differ in reducing capacity even when derived from the same tree.

Extraction method and solvent Aqueous, ethanolic, and methanolic extractions selectively recover different proportions of hydrophilic (ascorbic acid, chlorogenic acid) versus lipophilic (alpha-tocopherol, triterpenoids) compound classes. Most studies reviewed here ([22–25]) used aqueous extraction, but extraction temperature and time were not always reported in sufficient detail to allow cross-study comparison.

Storage conditions Phenolic and flavonoid compounds are susceptible to oxidative degradation during storage. Extracts prepared and used promptly would be expected to retain higher reducing activity than those stored for extended periods; however, none of the *Z. jujuba* studies reviewed report systematic stability data for the extract itself.

These sources of variability translate directly into the reproducibility of synthesis outcomes discussed in Sects. 4 and 5. The wide range of reported particle sizes for *Z. jujuba*-ZnO NPs (25–90 nm, Sect. 4.5) may reflect, in part, differences in extract phytochemical composition between studies in addition to differences in synthesis route and calcination temperature although the relative contribution of each factor cannot be determined from the published data because extract phytochemical characterization (e.g., total phenolic content, total flavonoid content) was not consistently reported alongside nanoparticle characterization data. We recommend that future studies report total phenolic content (TPC, as gallic acid equivalents) and total flavonoid content (TFC, as quercetin equivalents) for each extract batch used, enabling correlation between extract composition and resulting nanoparticle properties. The standardization implications of this variability are discussed further in Sect. 5.3.

3 Green synthesis mechanisms

3.1 Silver nanoparticles: phytochemical reduction mechanism

Synthesis of AgNPs by plant extract-mediated routes is generally described as a three-stage phytochemical reduction process [15, 30]. In Stage I, free Ag^+ ions in AgNO_3 solution form complexes with electron-rich functional groups of phytochemicals predominantly $-\text{OH}$ groups of flavonoids and carboxylate groups of phenolic acids forming transient Ag^+ –phytochemical complexes [16]. In Stage II, phenolic $-\text{OH}$ groups undergo oxidation to quinone structures via a two-electron transfer process, reducing coordinated Ag^+ to Ag^0 [17, 30]. Nuclei formed early in the reaction are thought to act as templates for further Ag^+ reduction at the particle surface, supporting progressive particle growth [31]. In Stage III, oxidized phytochemical species (quinones, carboxylates, polysaccharide chains) adsorb onto the expanding AgNP surface, contributing steric and electrostatic barriers that limit further growth and provide a surface charge that stabilizes the colloidal suspension [15, 32]. The formation of AgNPs is commonly associated with the appearance of a surface plasmon resonance (SPR) band in the UV-Vis spectrum at $\lambda_{\text{max}} = 420\text{--}450$ nm, attributed to the collective oscillatory response of surface conduction electrons to incident electromagnetic radiation [33, 34]. In the *Z. jujuba* system, Gavade et al. [22] reported $\lambda_{\text{max}} = 434$ nm, and Yuan et al. [23] (corrected citation; see Sect. 13) reported $\lambda_{\text{max}} = 446$ nm, both consistent with spherical AgNPs of 10–30 nm diameter and FCC crystal structure [22, 23]. A color change of the reaction medium from pale yellow to reddish-brown, observable within minutes under optimal conditions, provides a visual indicator supported quantitatively by the emergence of the SPR band [22]. The full width at half maximum (FWHM) of the SPR band is

related to particle size distribution: a narrow, symmetric band is associated with narrower size distribution and higher colloidal monodispersity [23, 35]. Figure 1 presents a schematic of the phyto-reduction mechanism for green synthesis of AgNPs using *Z. jujuba* leaf extract.

3.2 Zinc oxide nanoparticles: bio-mediated co-precipitation and thermal decomposition

The green synthesis of ZnO NPs is generally described as a three-step process: bio-mediated chelation of Zn^{2+} , precipitation of the resulting complexes as $\text{Zn}(\text{OH})_2$ upon addition of alkali, and conversion to crystalline ZnO via calcination [36, 37]. In Stage I, phytochemicals from *Z. jujuba* extract — polyphenols, polysaccharides, and carboxylic acid-bearing phenolics with multiple coordination sites — form chelate complexes with the $\text{Zn}(\text{NO}_3)_2$ precursor [38]. These phytochemical– Zn^{2+} complexes are proposed to influence final particle size and shape through spatial control of zinc ions prior to precipitation [36]. In Stage II, dropwise addition of NaOH to pH 10–11 precipitates $\text{Zn}(\text{OH})_2$ as a white gelatinous solid [37, 39]. The phytochemical coating on precipitating $\text{Zn}(\text{OH})_2$ particles is proposed to limit grain growth and agglomeration relative to chemically synthesized $\text{Zn}(\text{OH})_2$ [38, 40]. In Stage III (calcination), heating to 400–500 °C decomposes $\text{Zn}(\text{OH})_2$ to ZnO and H_2O via $\text{Zn}(\text{OH})_2 \rightarrow \text{ZnO} + \text{H}_2\text{O}$ [25, 36, 41], yielding the hexagonal wurtzite structure of ZnO (space group $\text{P6}_3\text{mc}$, JCPDS Card No.

36-1451). Organic residues from the phytochemical capping layer are removed during calcination, yielding ZnO powder of relatively high purity [37]. Figure 2 presents a process flow diagram for the green synthesis of ZnO NPs from *Z. jujuba* leaf extract via co-precipitation and calcination.

4 Influence of synthesis parameters on nanoparticle properties

The physicochemical properties of green-synthesized AgNPs and ZnO NPs size, shape, crystal structure, optical band gap, and colloidal stability depend on synthesis conditions. Systematic parameter optimization is therefore necessary to obtain reproducible nanoparticle properties for application. The following subsections examine each parameter using primary experimental data from the reviewed literature, with emphasis on publications from 2022 to 2025 and direct comparison of reported values where applicable.

4.1 Precursor concentration

Several recent studies [4, 22, 23, 42] have reported 1 mM AgNO_3 as the optimum concentration for AgNP synthesis. At this concentration, the molar ratio of phytochemical reducing equivalents to Ag^+ ions is reported to be sufficient for complete reduction and surface passivation. Liaquat et al. [42] demonstrated that AgNPs produced at 1 mM AgNO_3 with plant extract at 75 °C for

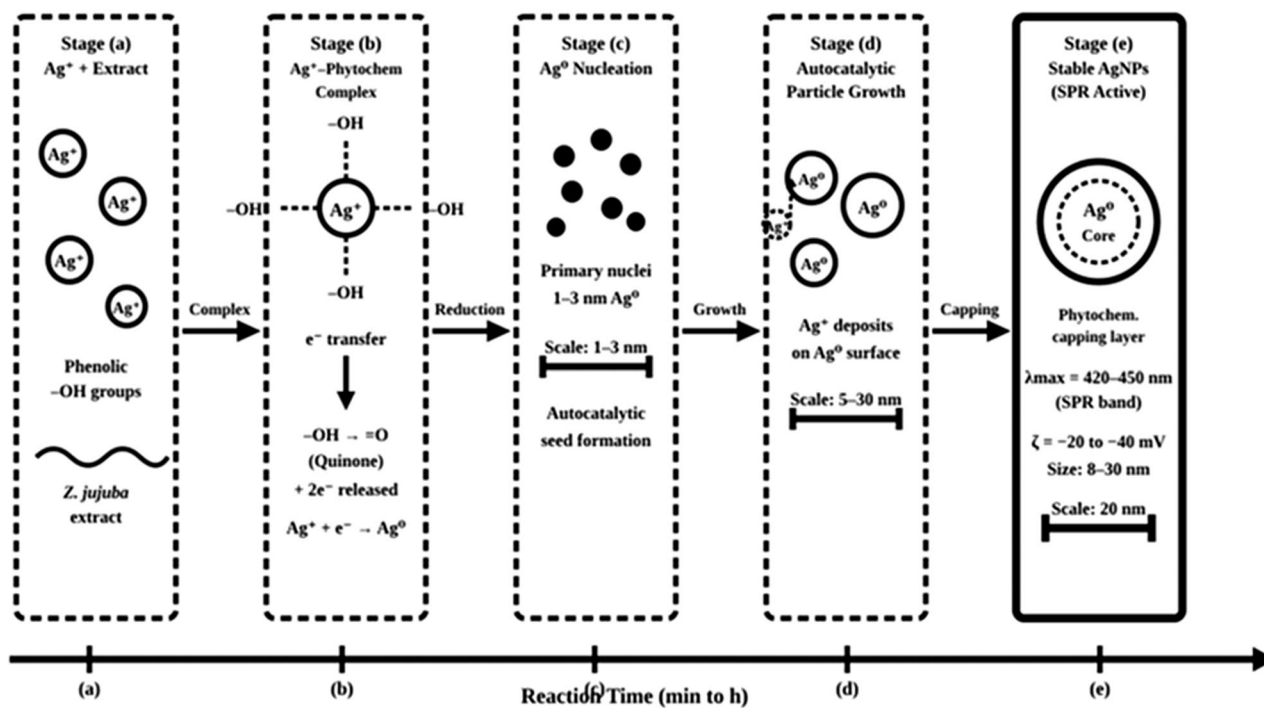


Fig. 1 Schematic representation of the phyto-reduction mechanism for green synthesis of silver nanoparticles (AgNPs) using *Ziziphus jujuba* leaf extract. The time-resolved sequence corresponds to: (a) Ag^+ ion complexation with flavonoid hydroxyl groups; (b–c) electron transfer and formation of Ag^0 nuclei; (d) autocatalytic particle growth; and (e) surface passivation by phytochemicals, yielding colloidal stable AgNPs with $\lambda_{\text{max}} = 420\text{--}450\text{ nm}$

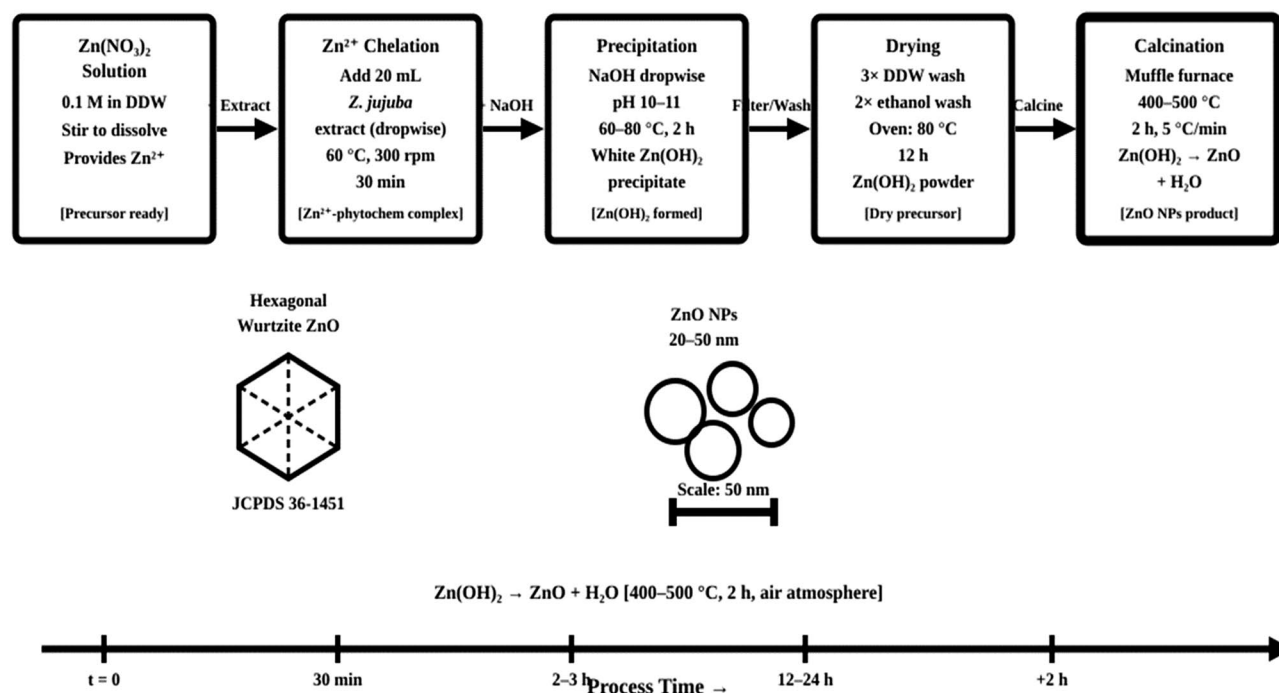


Fig. 2 Process flow diagram for the green synthesis of zinc oxide nanoparticles (ZnO NPs) via co-precipitation and calcination using *Ziziphus jujuba* leaf extract, annotated with process conditions reported in the reviewed literature

60 min exhibited a narrow SPR band and low polydispersity index (PDI), whereas higher concentrations yielded broader SPR bands and elevated PDI values consistent with particle agglomeration. Hassan et al. [43] reported FCC-structured AgNPs of 8 ± 1 nm at 1 mM using *Ziziphus* plant extract, a result broadly consistent with Liaqat et al. [42], although the two studies used different plant extracts and heating methods (conventional at 75 °C versus microwave-assisted synthesis), which limits direct mechanistic comparison. It should be noted that this comparison rests on only two studies using *Ziziphus* extracts at 1 mM, and broader conclusions about optimum concentration should be treated with caution pending more data from the *Z. jujuba* system specifically. For ZnO NPs, 0.1 M $\text{Zn(NO}_3)_2$ has been used across several 2022–2025 studies [25, 36, 37, 40, 41]; lower concentrations are associated with reduced precipitate yield, while higher concentrations are linked to rapid, less-controlled Zn(OH)_2 nucleation and larger aggregates after calcination [44].

4.2 Extract volume and phytochemical load

The ratio of phytochemical reducing/capping equivalents to metal ions governed by the volume of plant extract added to the precursor solution affects nucleation density, growth kinetics, and surface passivation efficiency [16, 17]. For *Ziziphus*-mediated AgNP synthesis, a 9:1 precursor-to-extract ratio (10% v/v extract) has been used [22, 23]. Larger extract volumes have been associated

with broader size distributions, possibly due to excess organic material at the particle surface, while smaller volumes are associated with larger and less stable particles [45]. However, these observations are based on studies using different plant species and synthesis conditions; a systematic dose-response study varying extract volume alone in the *Z. jujuba* system has not been reported, so direct conclusions for this extract are not yet possible. A higher extract volume (20% v/v) has been used for ZnO NP synthesis compared with AgNP synthesis (10% v/v) and has been attributed to the additional chelation demand of co-precipitation relative to simple reduction [21, 24, 25], although this attribution has not been tested by direct comparison at matched extract volumes.

4.3 pH of reaction mixture

pH affects nanoparticle synthesis through two mechanisms: it determines the ionization state of phytochemical reducing agents deprotonation of –OH groups above pH 8–9 is reported to increase electron-donating capacity and it influences the surface charge of nascent nanoparticles, with higher pH associated with more negative surface charge and electrostatic stabilisation [42, 46]. A pH range of 8–10 has been reported as suitable for AgNPs prepared from *Z. jujuba* and related plant extracts [22, 42]. Liaqat et al. [42] reported that the smallest particle sizes and narrowest SPR bands were obtained at neutral to mildly alkaline conditions (pH 7–9) in their optimization study. For ZnO NPs, complete and uniform

$\text{Zn}(\text{OH})_2$ precipitation is reported to require pH 10–11; below pH 9, precipitation is reported as incomplete, and above pH 12, $\text{Zn}(\text{OH})_2$ may re-dissolve as zincate ions ($\text{Zn}(\text{OH})_4^{2-}$) due to its amphoteric character, resulting in lower precipitate mass.

4.4 Reaction temperature

The kinetics of phytochemical oxidation (for AgNPs) and of Zn^{2+} chelation and precipitation (for ZnO NPs) are temperature-dependent. For AgNPs, the oxidation-reduction reaction proceeds more readily at elevated temperatures (60–80 °C), and smaller, more monodisperse particles have been associated with faster, more uniform nucleation under these conditions [42, 47]. Liaqat et al. [42] reported the narrowest SPR absorption band and smallest particle size at 75 °C with 60 min incubation. Hassan et al. [43] synthesized *Z. jujuba* fruit extract-mediated AgNPs by a microwave-assisted method, reporting a particle size of 8 ± 1 nm and zeta potential of -21.2 mV; the authors of that study attributed the small particle size to rapid and uniform microwave heating, although this attribution was not isolated from other differences between the microwave and conventional-heating protocols (e.g., total reaction time, extract concentration). Maturation temperatures of 60–80 °C during precipitation have been associated with improved crystallinity and reduced crystallite size of the wurtzite phase after calcination for ZnO NPs [36, 40].

4.5 Calcination temperature: a critical variable for ZnO NPs

Calcination temperature is widely identified as the parameter with the greatest influence on phase purity, crystallite size, specific surface area, optical band gap, and consequently antimicrobial and photocatalytic performance of ZnO NPs [36, 41, 44]. Calcination below 300 °C is reported to leave residual hydroxide phases,

visible as additional XRD reflections, together with organic residues from the phytochemical capping layer [44, 48].

Across the three available *Z. jujuba*-specific ZnO NP studies [24, 25, 27] a limited dataset from which only preliminary conclusions can be drawn, Alharthi et al. [24] reported ZnO NPs of 25 nm with an optical band gap of 2.70 eV and a specific surface area of 11.4 m²/g using microwave-assisted synthesis; the same research group (Alharthi et al. [25]) reported larger particles (90 ± 10 nm) using ultrasound-assisted synthesis, without reporting a systematic calcination-temperature series for that preparation. The roughly 3.6-fold difference in particle size between these two studies cannot be attributed definitively to any single factor calcination conditions, synthesis route, extract batch variation, or their combination -based on the published data alone. Chitrarasu et al. [27] reported *Z. jujuba*-ZnO NPs of 28.7 nm, an intermediate size, further illustrating the variability within this small body of work. Systematic studies on other plant-mediated ZnO NP systems confirm that crystallite size, surface area, and photocatalytic efficiency vary predictably with calcination temperature over the range 300–600 °C [41, 48], which supports the case for calcination-temperature optimization specific to *Z. jujuba*-ZnO NPs (see Sect. 11.3). Table 2 summarizes the critical synthesis parameters for plant-mediated AgNPs and ZnO NPs.

5 Critical evaluation of existing studies

5.1 Experimental limitations and reproducibility concerns

A recurring limitation across the *Z. jujuba* studies reviewed is incomplete reporting of experimental conditions necessary for replication. Extraction temperature and duration, extract-to-water ratio during preparation, and the age of the extract at the time of use are not consistently reported across studies [22–25]. Where particle

Table 2 Summary of critical synthesis parameters for plant-mediated AgNPs and ZnO NPs, with reported optimal conditions and consequences of parameter deviation. All temperatures in °C; all concentrations in M or mM

Parameter	AgNPs (reported optimum)	ZnO NPs (reported optimum)	Effect of Deviation	Key References (2022–2025)
Precursor concentration	1 mM AgNO ₃	0.1 M Zn(NO ₃) ₂ ·6 H ₂ O	Higher: agglomeration, high PDI. Lower: incomplete reaction, low yield	Liaqat et al. [42]; Hassan et al. [43]
Extract volume ratio	10% v/v	20% v/v	Insufficient: poor capping. Excess: possible organic contamination	Gavade et al. [22]; Alharthi et al. [25]*
pH	8–10	10–11	Low pH: slower/incomplete. Above pH 12 (ZnO): Zn(OH) ₂ re-dissolution	Liaqat et al. [42]; Sedefoglu [48]
Reaction temperature	60–80 °C	60–80 °C (precipitation stage)	Lower T: slower kinetics, larger polydisperse particles	Liaqat et al. [42]; Alharthi et al. [24]
Calcination temperature	Not applicable	400–500 °C (commonly reported)	< 300 °C: incomplete ZnO. >600 °C: grain coarsening, reduced surface area	Rusli et al. [41]; Sedefoglu [48]
Reaction/maturation time	1–4 h	2 h precipitation + up to 12 h ageing	Short: incomplete. Very long: Ostwald ripening, aggregation	Yuan et al. [23]*; Alharthi et al. [24]
Stirring speed	300–400 rpm	300 rpm	Insufficient mixing: local pH/concentration gradients, heterogeneous particles	Gavade et al. [22]; Alharthi et al. [25]*

size differs between studies for example, 11.5 ± 4.8 nm for AgNPs reported by Yuan et al. [23] versus 8 ± 1 nm by Hassan et al. [43], differences in extract preparation, precursor source, or measurement technique (HRTEM/AFM versus TEM) may all contribute, but the published methods sections do not provide sufficient detail to isolate the dominant cause.

The two available *Z. jujuba*-ZnO NP studies that report particle size ($n=2$; excluding the corrected lead-author study [27] which reports 28.7 nm) differ in synthesis route (microwave- versus ultrasound-assisted) and, where reported, in calcination conditions. Without a controlled comparison where only one variable differs at a time, the source of the 3.6-fold size difference (25 nm versus 90 ± 10 nm) between references [24] and [25] cannot be established from the existing data. These comparisons should therefore be regarded as preliminary observations rather than systematic findings.

5.2 Inconsistencies in characterization methods

Characterization techniques applied across the reviewed studies are broadly similar, but reporting depth varies. Scherrer crystallite size calculations are reported with different values for the shape constant k (commonly 0.89 or 0.94) across studies without stating the value used in some cases, introducing a difference of several percent in the reported crystallite size for the same diffraction data. Similarly, Tauc-method band gap values are reported without consistent specification of whether a direct or indirect transition model ($n=2$ or $n=1/2$) was assumed, affecting the extracted E_g value. Future *Z. jujuba* studies should explicitly state the Scherrer shape constant and the Tauc transition model, and should report measurement temperature and humidity for DLS/zeta potential measurements, to facilitate cross-study comparison.

5.3 Lack of standardization in extract reporting

As discussed in Sect. 2.3, none of the *Z. jujuba* nanoparticle synthesis studies reviewed [22–25] report total phenolic content (TPC) or total flavonoid content (TFC) for the extract batch used. Without this information, the phytochemical strength of each study's extract cannot be compared, and extract-related variation cannot be separated from synthesis-parameter-related variation in the reported outcomes. This is the most significant practical barrier to cross-study reproducibility in this area, and addressing it requires only straightforward colorimetric assays (Folin-Ciocalteu for TPC, aluminum chloride for TFC) that are now standard practice in plant extract characterization.

6 Characterization of green-synthesized AgNPs and ZnO NPs

6.1 UV-Visible spectroscopy

UV-Vis spectroscopy is widely used as a non-destructive, real-time method for monitoring AgNP formation via the diagnostic surface plasmon resonance (SPR) band [33, 34]. The SPR band position ($\lambda_{\max}$) depends on particle size (red-shift with increasing size), shape (broader or split peaks for non-spherical morphologies), surface chemistry (shift with capping agent polarity), and dielectric properties of the surrounding medium [33]. For *Z. jujuba*-synthesized AgNPs, $\lambda_{\max}$ values of 434 nm [22] and 446 nm [23] have been reported, consistent with spherical particles of approximately 20–30 nm diameter. For ZnO NPs, the optical absorption edge from UV-Vis spectra is used to calculate the optical band gap (E_g) via the Tauc relation: $(\alpha h\nu)^n = A(h\nu - E_g)$, where α is the absorption coefficient, $h\nu$ is the photon energy, A is a constant, and $n=2$ for ZnO as a direct-band-gap semiconductor [49]. Alharthi et al. [24] reported $E_g = 2.70$ eV for *Z. jujuba*-ZnO NPs red-shifted relative to bulk ZnO (3.37 eV) with possible contributing factors including quantum confinement effects and defect states associated with the phytochemical capping layer [24, 50].

6.2 X-ray diffraction (XRD)

XRD provides phase identification and, via the Debye-Scherrer equation, an estimate of crystallite size: $D = k\lambda / (\beta \cos\theta)$, where k is the Scherrer shape constant (see Sect. 5.2), $\lambda = 1.5406$ Å for CuK α radiation, β is the FWHM of the diffraction peak in radians, and θ is the Bragg angle. For AgNPs, characteristic Bragg reflections at $2\theta = 38.1^\circ$ (111), 44.3° (200), 64.5° (220), and 77.4° (311) confirm the face-centered cubic (FCC) crystal structure (JCPDS Card No. 04-0783) [22]. For ZnO NPs, reflections at $2\theta \approx 31.8^\circ$ (100), 34.4° (002), 36.3° (101), and related planes confirm the hexagonal wurtzite structure (JCPDS Card No. 36-1451) [24, 25, 27]. Chitrarasu et al. [27] confirmed this structure for *Z. jujuba*-ZnO NPs with TEM particle sizes of 28.7 nm. The three available *Z. jujuba*-ZnO NP studies report crystallite sizes ranging from 25 to 90 nm [24, 25, 27] a range that, as noted in Sect. 5.1, cannot be attributed to any single variable on the basis of current data. Figure 3 presents representative XRD patterns for green-synthesized AgNPs (A) and ZnO NPs (B).

6.3 Fourier-transform infrared spectroscopy (FTIR)

FTIR spectroscopy is used to provide evidence of phytochemical surface functionalization of green-synthesized nanoparticles [16, 51]. Comparison of FTIR spectra of the bare *Z. jujuba* extract with the nanoparticle product shows changes in absorption bands associated with reducing and capping functional groups. Reported diagnostic bands include broad O–H stretching at

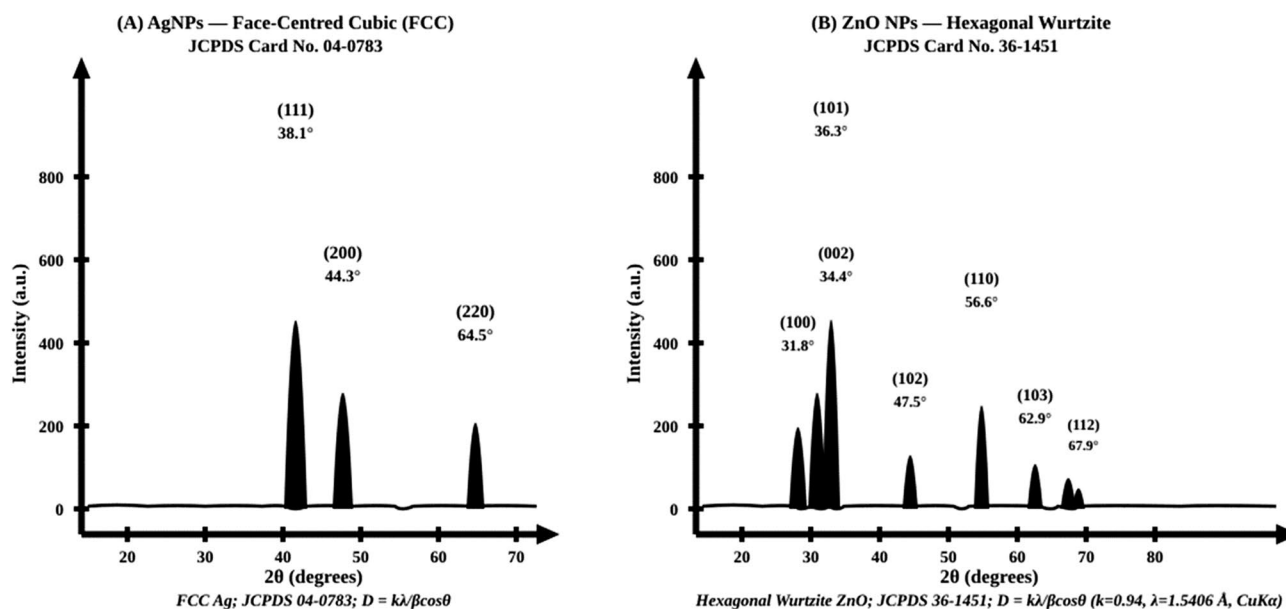


Fig. 3 Typical X-ray diffraction (XRD) patterns of green-synthesized **(A)** silver nanoparticles (AgNPs) with face-centered cubic (FCC) reflections indexed to JCPDS card number 04-0783 and **(B)** zinc oxide nanoparticles (ZnO NPs) with hexagonal wurtzite reflections indexed to JCPDS card number 36-1451. Key Miller indices are indicated. Crystallite size calculated using the Debye-Scherrer equation ($D = k\lambda/\beta\cos\theta$)

3200–3500 cm^{-1} ; C=O stretching at 1630–1750 cm^{-1} ; C–O stretching at 1050–1260 cm^{-1} ; and C=C aromatic stretching at 1450–1600 cm^{-1} [51, 52]. Shifts in these bands between the extract and nanoparticle spectra are interpreted as evidence that the corresponding functional groups are coordinated to, rather than only physically mixed with, the nanoparticle surface [22, 51]. For ZnO NPs, an additional absorption band at 400–600 cm^{-1} associated with Zn–O stretching confirms ZnO bond formation [53]. This band was reported at 520 cm^{-1} for ultrasound-assisted *Z. jujuba*-ZnO NPs [25] and at 468 cm^{-1} for the microwave-assisted product [24, 54] a difference of approximately 52 cm^{-1} that has not been discussed in either study but may reflect differences in particle size, defect density, or residual organic content between the two synthesis routes, as noted in the characterization inconsistencies discussed in Sect. 5.2.

6.4 Scanning electron microscopy, transmission electron microscopy, and energy-dispersive X-ray spectroscopy

SEM provides surface morphology information at the microscale, while TEM provides higher-resolution information on internal structure, particle size distribution, and lattice spacing via high-resolution TEM (HRTEM) and selected area electron diffraction (SAED) [55]. Spherical morphology of plant-mediated AgNPs from *Z. jujuba* has been reported by Yuan et al. [23] (11.5 ± 4.8 nm by HRTEM and AFM), Gavade et al. [22] (20–30 nm by TEM), and Hassan et al. [43] (8 ± 1 nm by TEM for microwave-assisted, fruit-extract-mediated AgNPs). These three studies form the available size dataset for *Z.*

jujuba-AgNPs; given this limited number, the size range of 8–30 nm across these reports should be considered a preliminary observation rather than a well-established characteristic.

6.5 Dynamic light scattering and zeta potential analysis

Dynamic Light Scattering (DLS) provides the hydrodynamic particle diameter and polydispersity index (PDI) of nanoparticles in the colloidal state, with PDI < 0.3 generally considered acceptable for pharmaceutical and sensing applications [56, 57]. DLS-measured hydrodynamic diameters are typically larger than TEM-measured sizes, attributed to the hydration shell of surface-bound phytochemicals [56]. The commonly used stability threshold is $|\zeta| \geq 25$ mV [57]. Reported ζ values for green-synthesized AgNPs are generally in the range –20 to –40 mV, attributed to deprotonated carboxylate and phenolate groups on the phytochemical surface layer [42, 58]. Hassan et al. [43] reported $\zeta = -21.2$ mV for *Z. jujuba* fruit-extract AgNPs, a value within this range but toward its lower (less stable) bound. These measurements apply to colloidal AgNP suspensions; ZnO NPs, isolated as a dry powder after calcination, are characterized by BET surface area analysis instead.

6.6 BET surface area, TGA/DSC, and photoluminescence

BET nitrogen adsorption provides the specific surface area (m^2/g) and pore size distribution of ZnO NPs, reported to relate to photocatalytic efficiency and antimicrobial contact area [24, 44]. Alharthi et al. [24] reported 11.4 m^2/g for microwave-assisted *Z. jujuba*-ZnO NPs,

Table 3 Summary of characterization techniques applied to green-synthesized AgNPs and ZnO NPs, key diagnostic signatures, and representative findings from *Ziziphus jujuba* studies

Technique	Information Obtained	AgNPs Diagnostic	ZnO NPs Diagnostic	Z. jujuba Literature
UV-Vis spectroscopy	SPR band; optical band gap	λ_{max} 420–450 nm; narrower band associated with greater monodispersity	Absorption edge $\rightarrow$ Eg 2.7–3.4 eV (Tauc method)	[22]: 434 nm; [23]: 446 nm; [24]: Eg 2.70 eV
XRD	Phase identity; crystal size (Scherrer)	FCC Ag; JCPDS 04-0783; $2\theta \approx 38.1^\circ$, 44.3° , 64.5° , 77.4°	Wurtzite ZnO; JCPDS 36-1451; $2\theta \approx 31.8^\circ$ – 69.1°	[22, 23]: FCC confirmed; [24, 25, 27]: wurtzite confirmed
FTIR	Surface functional groups; capping evidence	O–H (3200–3500), C=O (1630–1750), C–O (1050–1260) cm^{-1} band shifts	Zn–O band at 400–600 cm^{-1} ; O–H, C=O, C–O bands	[22, 24, 25]: phytochemical capping reported
SEM/TEM + EDX	Morphology; size; elemental composition	Spherical; 8–30 nm; Ag signal $\sim$ 3 keV	Spherical/hexagonal; 25–90 nm; Zn, O signals	[22]: 20–30 nm; [23]: 11.5 nm; [24]: 25 nm; [25]: 90 nm; [27]: 28.7 nm
DLS + Zeta potential	Hydrodynamic size; PDI; colloidal stability	PDI < 0.3 (where reported); $\zeta \approx -20$ to -40 mV	Not applicable (dry powder)	[43]: $\zeta = -21.2$ mV
BET surface area	Specific surface area; porosity	Not applicable (colloidal)	Reported range 10–30 m^2/g for green-synthesized ZnO	[24]: 11.4 m^2/g ; [25]: 14.23 m^2/g
TGA/DSC	Decomposition temperature; organic content	Not applicable	Identifies $\text{Zn}(\text{OH})_2 \rightarrow \text{ZnO}$ transition; informs calcination temperature selection	[41]: recommended for calcination design
PL spectroscopy	Defect states; oxygen vacancies	Not applicable	NBE UV emission; visible defect bands	[24]: characterized; [48]: PL analysed

and Alharthi et al. [25] reported 14.23 m^2/g for ultrasound-assisted synthesis. As noted in Sect. 5.1, the higher BET value for the ultrasound-assisted preparation is difficult to reconcile with its larger particle size (90 ± 10 nm), as larger particles would generally be expected to yield lower specific surface area — a possible inconsistency that highlights the need for the standardized characterization reporting discussed in Sect. 5.2. TGA and DSC of the dried $\text{Zn}(\text{OH})_2$ precursor before calcination can identify the decomposition temperature and organic residue content, supporting evidence-based calcination temperature selection [41, 44]. Near-band-edge (NBE) UV emission in photoluminescence (PL) spectra reflects intrinsic crystal quality, while visible-range defect emission bands (green: oxygen vacancy; yellow-orange: interstitial oxygen) inform the defect chemistry that influences photocatalytic activity [50, 54, 59]. Table 3 summarizes the characterization techniques applied to green-synthesized AgNPs and ZnO NPs.

7 Multifunctional applications

7.1 Antimicrobial activity

Antimicrobial activity is among the most widely reported and clinically relevant properties of green-synthesized AgNPs and ZnO NPs [3, 4, 7, 8]. Antimicrobial resistance (AMR) is recognized as a major global public health threat, with multidrug-resistant (MDR) bacteria implicated in an estimated 1.27 million deaths annually [60]. The multi-target mode of action proposed for green-synthesized nanoparticles disruption of cell membranes, generation of ROS, interference with DNA replication, and inhibition of respiratory chain enzymes is suggested to reduce the likelihood of resistance developing via single or limited mutations, compared with conventional single-target antibiotics [61–63]. Proposed bactericidal mechanisms of AgNPs include: (i) release of Ag^+ ions that bind thiol ($-\text{SH}$) groups in bacterial membrane proteins, disrupting membrane integrity and proton motive force; (ii) generation of superoxide ($\text{O}_2^{\bullet-}$) and hydroxyl ($\bullet\text{OH}$) radicals causing oxidative damage to membrane lipids, proteins, and DNA; (iii) direct interaction of AgNPs with bacterial DNA, potentially inhibiting replication; and (iv) inhibition of membrane-bound ATP synthase and respiratory chain cytochrome oxidases [61–63].

Z. jujuba-AgNPs have been reported to produce zone of inhibition (ZOI) values of 14–18 mm against *Escherichia coli* at 25–100 $\mu\text{g}/\text{mL}$ [22] and activity against five clinical MDR pathogens including *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Klebsiella pneumoniae* [26]. Hassan et al. [64] (2025) reported that *Ziziphus* plant-extract-mediated AgNPs showed antibacterial activity, with Gram-negative bacteria reported as more susceptible than Gram-positive bacteria, attributed by the authors to the thinner peptidoglycan cell wall of

Gram-negative organisms. Liaqat et al. [42] reported ZOI values of 16 mm against *Bacillus subtilis* and 17 ± 0.8 mm against *S. aureus*, comparable to standard antibiotic controls at the same concentration in that study.

While ZOI measurements provide a useful initial indicator of antimicrobial activity, minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) values are widely regarded as more clinically meaningful because they express activity in terms of the lowest effective concentration rather than a diffusion-dependent inhibition zone diameter [61]. Of the *Z. jujuba*-specific AgNP and ZnO NP studies reviewed [22–27], none reported MIC or MBC values a gap that limits direct comparison of antimicrobial potency across studies and with conventional antibiotics. In the broader green-synthesis literature, MIC values for plant-extract-mediated AgNPs have been reported in the range of 2–64 $\mu\text{g/mL}$ against *E. coli* and *S. aureus* [61, 65], and for ZnO NPs in the range of 62.5–500 $\mu\text{g/mL}$ depending on particle size and test organism [66]. These ranges are provided as reference benchmarks; future *Z. jujuba* nanoparticle studies should report MIC and MBC alongside ZOI to enable meaningful comparison with this literature.

Proposed antimicrobial mechanisms for ZnO NPs include: (i) ROS generation under UV light and, to a

lesser extent, in the dark due to surface oxygen vacancies; (ii) release of Zn^{2+} ions in aqueous media, proposed to inhibit ATPases; and (iii) direct membrane contact and disruption associated with the positive surface charge of ZnO at physiological pH [66, 67]. *Z. jujuba*-ZnO NPs have been reported to produce ZOI values of 15 mm against *E. coli* and 11 mm against *S. aureus* [25], evaluated using three assay methods — disc diffusion, column filtration, and aqueous suspension — reporting consistent activity across methods. Direct evidence for antibiofilm activity of *Z. jujuba* against a clinically relevant MDR pathogen is provided by Turkey and Nazeam [68], who reported that a standardized *Z. jujuba* seed fraction reduced biofilm formation and virulence gene expression in clinical *Pseudomonas aeruginosa* isolates. This study used the plant fraction directly rather than a nanoparticle formulation and is included here as supporting evidence for the antimicrobial relevance of *Z. jujuba* phytochemicals, distinct from evidence specific to *Z. jujuba*-derived nanoparticles. The quantitative comparison of antimicrobial performance data is presented in Table 4.

7.2 Photocatalytic activity

Degradation of persistent organic pollutants particularly synthetic dyes from the textile, paper, and pharmaceutical

Table 4 Quantitative comparison of multifunctional application performance data for AgNPs and ZnO NPs from *Ziziphus* species and reference plant systems

Application	NP System	Key Performance	Pathogen/Target	MIC/MBC	Evidence Type	Ref.
Antibacterial	<i>Z. jujuba</i> -AgNPs	ZOI 14–18 mm at 25–100 $\mu\text{g/mL}$	<i>E. coli</i> , <i>S. aureus</i> , <i>P. aeruginosa</i>	Not reported	Direct	[22]
Antibacterial	<i>Z. jujuba</i> -AgNPs (microwave)	Size 8 ± 1 nm; $\zeta = -21.2$ mV	<i>B. subtilis</i> , <i>S. aureus</i>	Not reported	Direct	[43]
Antibacterial	<i>Ziziphus</i> -AgNPs (related species)	ZOI vs. 5 MDR pathogens	<i>E. coli</i> , <i>S. aureus</i> , <i>K. pneumoniae</i>	Not reported	Indirect (related species)	[64]
Antibacterial	<i>Z. jujuba</i> -ZnO NPs	ZOI 15 mm (<i>E. coli</i>), 11 mm (<i>S. aureus</i>)	<i>E. coli</i> , <i>S. aureus</i>	Not reported	Direct	[25]
Antibiofilm	<i>Z. jujuba</i> seed fraction (not NP)	Reduced biofilm formation and virulence gene expression	<i>P. aeruginosa</i> (clinical isolates)	Not applicable (plant fraction)	Direct (extract, not NP)	[68]
Photocatalysis	<i>Z. jujuba</i> -ZnO NPs	Selective MO degradation; Eg 2.70 eV	Methyl orange, methylene blue	Not applicable	Direct	[24]
Photocatalysis	<i>Z. jujuba</i> -ZnO NPs	~90% Acid Black 1 degradation in 60 min, pH 7	Acid Black 1 dye	Not applicable	Direct	[27]
Anticancer	<i>Z. jujuba</i> -ZnO NPs	CCK-8 assay (HCT-116, HepG2); 20–600 $\mu\text{g/mL}$	Colon and liver cancer cell lines	Not applicable	Direct	[25]
Anticancer (extract)	<i>Z. jujuba</i> extract (not NP)	Reduced proliferation (colorectal cancer cells)	Colorectal cancer cell lines	Not applicable	Direct (extract, not NP)	[78]
Anticancer	ZnO–Ag NPs (other plant)	IC_{50} 30 $\mu\text{g/mL}$ (NCI-H460)	Lung cancer cells	Not applicable	Indirect	[81]
Antioxidant	Ag-ZnO NPs (other plant)	DPPH IC_{50} 85.67 $\mu\text{g/mL}$; ABTS IC_{50} 93.78 $\mu\text{g/mL}$	DPPH, ABTS radicals	Not applicable	Indirect	[76]
Biosensing	<i>Z. jujuba</i> -ZnO NPs	Demonstrated; LOD and other metrics not reported	Electrochemical analytes	Not applicable	Direct (metrics not reported)	[85]
Heavy metal sensing	Plant-extract AgNPs (other plant)	ppb detection Hg^{2+} , Pb^{2+} , Cu^{2+}	Heavy metals in water	Not applicable	Indirect	[58]

ZOI=zone of inhibition; IC_{50} = half-maximal inhibitory concentration; MIC=minimum inhibitory concentration; MBC=minimum bactericidal concentration. 'Direct' indicates the study used *Z. jujuba* extract; 'Indirect' indicates a related plant or nanoparticle system

industries is a widely studied application of ZnO NPs [69, 70]. The photocatalytic mechanism involves: (i) absorption of photons with energy $\geq E_g$ generating electron-hole (e^-/h^+) pairs; (ii) holes migrating to the particle surface, oxidizing surface hydroxide ions and adsorbed water to produce $\bullet OH$ radicals; (iii) electrons reducing dissolved O_2 to superoxide radical anions ($O_2^{\bullet -}$); and (iv) $\bullet OH$ and $O_2^{\bullet -}$ radicals mineralizing organic dye chromophores to CO_2 , H_2O , and inorganic ions [69–71]. Alharthi et al. [24] reported selective photocatalytic degradation of methyl orange (MO, anionic) over methylene blue (MB, cationic) under solar irradiation using *Z. jujuba*-ZnO NPs, attributed to electrostatic attraction between the anionic dye and the positively charged ZnO surface. Chitrarasu et al. [27] reported approximately 90% degradation of Acid Black 1 dye within 60 min under natural sunlight at pH 7, with *Z. jujuba*-ZnO NPs (28.7 nm) showing slightly different performance from *Syzygium jambolanum*-ZnO NPs (30.2 nm), indicating that plant extract source may influence photocatalytic activity, although the mechanistic basis for this difference was not established. Al-Zaban [21] also reported photocatalytic dye degradation using *Z. jujuba*-ZnO NPs with kinetic and thermodynamic characterization. Reported degradation efficiencies for optimized plant-mediated ZnO NPs against various dyes range from approximately 84–98% over 60–120 min under UV or solar irradiation [36, 72]. Figure 4 presents

an energy band diagram for the photocatalytic mechanism of green-synthesized ZnO NPs.

7.3 Antioxidant activity

Free radical scavenging activity of plant-extract-mediated AgNPs and ZnO NPs has been reported to exceed that of the corresponding bare plant extract in some studies [73, 74], attributed to the combined contribution of the phytochemical surface layer (DPPH scavenging by flavonoids and phenolics) and the electron-accepting capacity of the nanoparticle core [73]. Antioxidant capacity is typically measured using DPPH (2,2-diphenyl-1-picrylhydrazyl), ABTS (2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid)), and FRAP (ferric reducing antioxidant power) radical-scavenging assays, reported as IC_{50} values [74]. Direct DPPH/ABTS data for *Z. jujuba*-AgNPs and *Z. jujuba*-ZnO NPs have not been identified in the literature reviewed. By way of indirect comparison, IC_{50} values in the range 50–150 $\mu g/mL$ have been reported for green-synthesized AgNPs from other plant systems [73, 75], and ZnO–Ag bimetallic nanoparticles synthesized from *Andrographis macrobotrys* were reported with DPPH IC_{50} = 85.67 $\mu g/mL$ and ABTS IC_{50} = 93.78 $\mu g/mL$ [76, 77]. These values are presented as context for the antioxidant potential achievable from phytochemically rich plant-extract systems generally; they should not be interpreted as representative of *Z. jujuba*-specific

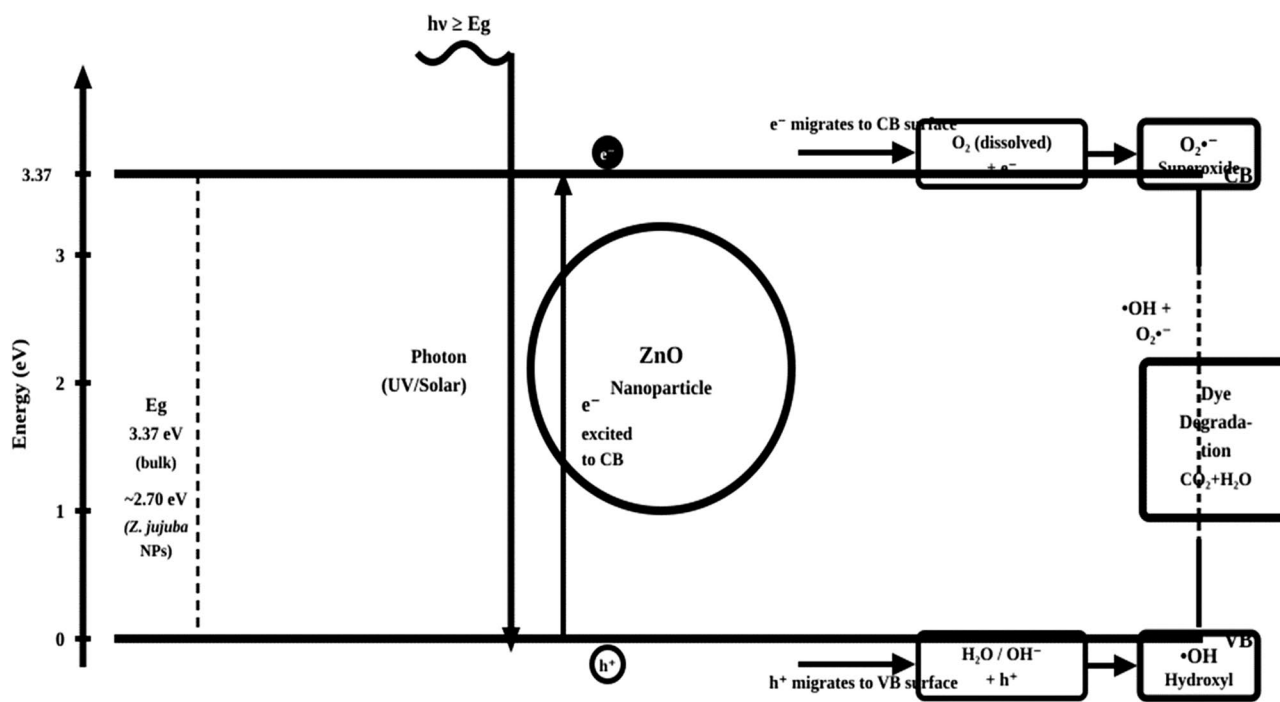


Fig. 4 Energy band diagram for the photocatalytic mechanism of green-synthesized ZnO nanoparticles. Photoexcitation by UV or visible light ($h\nu \geq E_g$) generates electron-hole pairs; conduction-band electrons reduce O_2 to superoxide ($O_2^{\bullet -}$), while valence-band holes oxidize water/ OH^- to hydroxyl radicals ($\bullet OH$). Both radical species participate in the oxidative degradation of organic dye pollutants

nanoparticles, for which no direct antioxidant assay data are currently available.

7.4 Anticancer and cytotoxic activity

7.4.1 Direct evidence from *Z. jujuba* systems

Two sources of direct evidence are available for *Z. jujuba* in relation to anticancer activity. First, Alharthi et al. [25] reported that ultrasound-assisted *Z. jujuba*-ZnO NPs were evaluated for cytotoxicity against HCT-116 (colon cancer) and HepG2 (liver cancer) cell lines using a cell counting kit-8 (CCK-8) assay, with concentration-dependent reduction in cell viability and morphological changes in HCT-116 cells reported after 24 h exposure at concentrations of 20–600 µg/mL [25]. This is, to our knowledge, the only published cytotoxicity dataset for *Z. jujuba*-derived nanoparticles against human cancer cell lines. Second, Pourali et al. [78] reported an in vitro study of *Z. jujuba* extract not a nanoparticle formulation against colorectal cancer cell lines, finding that the extract reduced cell proliferation in a concentration-dependent manner. This study provides supporting rationale for testing *Z. jujuba*-derived nanoparticles against the same colorectal cancer cell lines, but does not itself constitute evidence for the nanoparticle systems discussed in this review.

7.4.2 Indirect evidence from related nanoparticle systems

Beyond the *Z. jujuba*-specific evidence above, ZnO NPs and Ag-ZnO nanocomposites synthesized from other plant extracts have been reported as selective anticancer agents, with proposed mechanisms involving preferential ROS generation in cancer cells leading to mitochondrial membrane depolarization, cytochrome c release, and caspase-3-mediated apoptosis [79, 80]. The following findings relate to non-*Z. jujuba* systems and are presented as context for the mechanisms that might apply to *Z. jujuba*-derived nanoparticles, not as evidence for *Z. jujuba* itself. Bimetallic ZnO–Ag NPs prepared from *Trichosanthes dioica* seed extract were reported with an IC₅₀ of 30 µg/mL against NCI-H460 lung cancer cells [81]. Hassan et al. [64] (2025) reported that *Ziziphus*-mediated AgNPs from a related *Ziziphus* species, not *Z. jujuba* specifically induced apoptotic changes in cancer cell lines [64, 82]. In vitro cytotoxicity profiling using standardized assays is increasingly expected for biomedical nanoparticle publications [83, 84], and for *Z. jujuba*-AgNPs specifically, no cytotoxicity data against human cell lines were identified in the literature reviewed.

7.5 Biosensing applications

7.5.1 Demonstrated applications

Wahid et al. [85] synthesized ZnO NPs from *Z. jujuba* leaves and reported their use in a biosensing context. Based on the information available in the source

reviewed, the specific analyte detected, the limit of detection (LOD), sensitivity, selectivity, and response time of this sensor were not reported in sufficient detail to summarize quantitatively [86, 87]. The study is therefore noted as a demonstrated application of *Z. jujuba*-ZnO NPs in a biosensing format, without further quantitative claims. Outside the *Z. jujuba* system, Saber et al. [58] reported colorimetric detection of Hg²⁺, Pb²⁺, and Cu²⁺ ions at part-per-billion (ppb) concentrations using plant-extract-mediated AgNPs, demonstrating the principle of colorimetric heavy-metal sensing using green-synthesized AgNPs in general but not using *Z. jujuba* extract.

7.5.2 Proposed (not yet demonstrated) applications

The strong and tuneable SPR response of AgNPs is, in principle, suited to colorimetric and surface-enhanced Raman scattering (SERS)-based detection of trace analytes, where analyte-induced aggregation produces a measurable shift or quenching of the SPR band [88, 89]. Similarly, the surface oxygen vacancies and band gap of ZnO NPs are, in principle, compatible with electrochemical biosensor transduction. However, neither application configuration has been demonstrated specifically for *Z. jujuba*-AgNPs or *Z. jujuba*-ZnO NPs with reported sensor performance metrics [90, 91]. Future studies reporting biosensing claims for *Z. jujuba*-derived nanoparticles should report LOD, sensitivity, selectivity, linear dynamic range, and response time. Table 4 presents a quantitative comparison of multifunctional application performance data.

8 Comparative analysis of literature: AgNPs versus ZnO NPs from *Ziziphus* systems

Table 5 presents a comparison of primary studies using *Ziziphus* species and reference plant systems for AgNP and ZnO NP green synthesis, with corrected author attributions applied throughout (Sect. 13) and a new column summarizing the principal limitation of each study.

Examination of Table 5 indicates the following patterns. First, *Z. jujuba*-mediated AgNPs are reported with particle sizes between 8 and 30 nm across the studies identified [22, 23, 43], though direct comparison is limited by differences in extract preparation and measurement technique (Sect. 5.1). Second, *Z. jujuba*-ZnO NPs show a roughly 3.6-fold range in reported size (25–90 nm) across three studies [24, 25, 27], with the calcination conditions and synthesis route differing between studies (Sect. 4.5, 5.1). Third, no study in Table 5 synthesizes both AgNPs and ZnO NPs from the same *Z. jujuba* extract batch, a methodological gap discussed further in Sect. 11. Fourth, direct evidence for biosensing applications of *Z. jujuba*-derived nanoparticles is limited to a single study without reported performance metrics (Sect. 7.5). Fifth, none of the *Z. jujuba*-specific studies reviewed reported MIC

Table 5 Comparison of primary green synthesis studies using *Ziziphus* species and reference plant systems, with corrected author attributions (Sect. 13) and principal study limitations

Study / Year	Plant / Part	NP Type	Size (nm)	Synthesis Route	Application	Journal	Study limitation
Gavade et al., 2015 [22]	<i>Z. jujuba</i> / Leaf	AgNPs	20–30	Aqueous, RT	Antibacterial, antifungal	Spectrochim. Acta A	No MIC/MBC data; no extract TPC/TFC reported
Yuan et al., 2017 [23]*	<i>Z. jujuba</i> cv. Dongzao / Leaf	AgNPs	11.5 ± 4.8	Aqueous, 25 °C	Antibacterial, catalytic	IET Nanobiotechnol.	Author list corrected (see Sect. 13); no MIC/MBC; no extract characterisation
Alharthi et al., 2021 [24]	<i>Z. jujuba</i> / Leaf	ZnO NPs	25	Microwave-assisted	Photocatalysis	Nanomaterials	No calcination T series; no ecotoxicity; no cytotoxicity
Alharthi et al., 2023 [25]*	<i>Z. jujuba</i> / Leaf	ZnO NPs	90 ± 10	Ultrasound-assisted	Antibacterial, anticancer	Separations	Author list corrected; no calcination T series; BET/size inconsistency (Sect. 6.6)
Al-Zaban et al., 2021 [21]*	<i>Z. jujuba</i> / Leaf	ZnO NPs	Variable	Aqueous co-precip.	Dye degradation	J Phys Chem Solids	Journal corrected (see Sect. 13); no particle size distribution reported
Al-Askar et al., 2022 [26]	<i>Z. spinachristi</i> / Leaf	AgNPs	24 ± 4	Aqueous, RT	5-pathogen MDR antibacterial	Herald Open Access	Related <i>Ziziphus</i> species (not <i>Z. jujuba</i>); no MIC/MBC
Hassan et al., 2025 [64]	<i>Ziziphus</i> (related) / Leaf and Stem	AgNPs	5.77–24	Aqueous, 70–80 °C	Antibacterial, anticancer	J Appl Organometal Chem	Related species (not <i>Z. jujuba</i>); species not fully specified
Saber et al., 2024 [58]	<i>Vachellia tortilis</i> / Leaf	AgNPs	20–30	Aqueous	Antibacterial, heavy metal sensing	BMC Chem	Non- <i>Ziziphus</i> system; indirect evidence only
Rusli et al., 2024 [41]	<i>Garcinia mangostana</i> / Leaf	ZnO NPs	40.98–46.92	Co-precip., T series	Calcination T optimisation	J Adv Vet Anim Res	Non- <i>Ziziphus</i> system; no antimicrobial data
Sedefoglu, 2024 [48]	<i>Myrtus communis</i> / Leaf	ZnO NPs	Variable	Co-precipitation	Photocatalysis (~90% MB)	Ceram Int	Non- <i>Ziziphus</i> system; indirect evidence only
Chitrarasu et al., 2025 [27]*	<i>Z. jujuba</i> + <i>S. jambolana</i> / Leaf	ZnO NPs	28.7	Aqueous co-precip.	~90% Acid Black 1 degradation (60 min)	ChemistrySelect	Lead author corrected (see Sect. 13); no cytotoxicity or MIC/MBC
Turkey & Nazeam, 2025 [68]	<i>Z. jujuba</i> seed (not NP)	N/A (plant fraction)	N/A	Solvent fractionation	Antibiofilm vs. <i>P. aeruginosa</i>	AMB Express	Plant extract study; no nanoparticle synthesis; indirect evidence only

or MBC values (Sect. 7.1, Table 4), which substantially limits the comparability of antimicrobial findings across studies [92, 93].

The comparative picture emerging from Table 5a small number of *Z. jujuba*-specific studies with substantial variability in reported properties and limited application data naturally raises two sets of practical questions that extend beyond the immediate synthesis and characterization literature. The first concerns whether the synthesis routes described here could be transferred to larger production scales and whether the green chemistry advantages claimed at laboratory scale would be retained or diminished at pilot or industrial scale (Sect. 9). The second concerns the environmental fate and safety of the resulting nanoparticles: avoiding hazardous synthesis reagents does not in itself guarantee that the product nanoparticles are environmentally benign, and

this question has received no attention in the *Z. jujuba*-specific literature to date (Sect. 10).

9 Scale-Up and industrial perspective

All studies reviewed in this paper were conducted at laboratory scale, typically using reaction volumes of 100–500 mL and extract volumes of 10–40 mL [22–25, 42, 43]. None of the *Z. jujuba* studies reviewed report scale-up data, batch-to-batch consistency data across multiple production runs, or cost analyses comparing green synthesis with conventional chemical synthesis routes. Several considerations relevant to potential scale-up can nonetheless be identified from the broader green-synthesis literature [11, 12].

Extract supply and standardization. Scale-up would require a consistent and sufficient supply of *Z. jujuba* leaves with characterized phytochemical content (Sect.

2.3 and 5.3). Seasonal and geographical variability in leaf composition would need to be addressed through standardized sourcing, batch testing (TPC/TFC assays), or blending of multiple harvests to reduce batch-to-batch variation. Reaction vessel design and mixing. Laboratory protocols rely on magnetic stirring at 300–400 rpm in small vessels (Sect. 4.2). At pilot or production scale, achieving equivalent local mixing and heat transfer for the 60–80 °C reaction temperatures reported in Sect. 4.4 would require appropriately designed stirred-tank reactors. Whether the reported nanoparticle sizes and yields are maintained under different mixing and heat-transfer regimes has not been investigated for *Z. jujuba* systems.

Calcination at scale (ZnO NPs). The calcination step (400–500 °C, Sect. 4.5) is energy-intensive. At production scale, this represents a recurring operating cost that would need to be weighed against the environmental benefits of avoiding chemical reducing agents. Energy consumption data for the calcination step were not reported in any of the studies reviewed.

Cost comparison. No cost data comparing *Z. jujuba*-extract-mediated synthesis with conventional chemical synthesis were identified. A meaningful comparison would need to account for extract preparation, reagent costs (AgNO₃, Zn(NO₃)₂, NaOH), and energy costs on both sides. In our assessment, *Z. jujuba*-mediated synthesis of AgNPs and ZnO NPs remains at an early, laboratory-scale stage of development, and claims regarding industrial feasibility or cost-effectiveness would not currently be supported by the published evidence [94, 95].

10 Environmental safety and ecotoxicity considerations

Green synthesis is often presented as an environmentally preferable alternative to chemical synthesis because it avoids hazardous reducing agents such as sodium borohydride and hydrazine (Sect. 1) [96, 97]. However, this does not imply that the resulting nanoparticles are without environmental or toxicological concern. AgNPs and ZnO NPs, regardless of synthesis route, can be toxic to aquatic and terrestrial organisms at sufficient concentrations [73, 98]. Reported mechanisms of nanoparticle ecotoxicity include release of Ag⁺ or Zn²⁺ ions, generation of ROS in exposed organisms, and physical interactions between nanoparticles and biological membranes [73, 98].

None of the *Z. jujuba*-AgNP or *Z. jujuba*-ZnO NP studies reviewed [22–25, 43] report ecotoxicological assessment for example, acute toxicity testing against standard model organisms such as *Daphnia magna*, zebrafish embryos, or brine shrimp (*Artemia salina*), which are commonly used in nanoparticle ecotoxicity

studies [73]. The phytochemical capping layer on green-synthesized nanoparticles has been proposed in some studies to reduce ion release rates relative to uncapped nanoparticles, potentially lowering acute toxicity; however, this proposed benefit has not been directly tested for *Z. jujuba*-derived nanoparticles.

For applications discussed in Sect. 7 specifically photocatalytic dye degradation (Sect. 7.2) and biosensing for heavy-metal detection (Sect. 7.5) that involve use of nanoparticles in aqueous environments, assessment of nanoparticle recovery after the treatment process [99], toxicity of degradation by-products, and ecotoxicity of the nanoparticles themselves would be necessary before field application could be considered. We recommend that future *Z. jujuba* nanoparticle studies intended for environmental or biomedical application include, at minimum, an acute toxicity assay against one standard model organism appropriate to the intended application route.

11 Research Gaps and Recommended Future Directions

11.1 Theme 1: Synthesis Standardization and Reporting

Gap: Extraction conditions and extract phytochemical content (TPC, TFC) are not consistently reported across *Z. jujuba* studies (Sect. 2.3, 5.3), limiting reproducibility and cross-study comparison [100, 101].

Recommendation: Future studies should report extraction temperature, duration, solvent-to-leaf ratio, extract storage conditions, and TPC/TFC values for each extract batch.

Suggested approach: A standardized reporting checklist for plant-mediated nanoparticle synthesis studies, analogous to reporting guidelines in other fields, would address this at the community level.

11.2 Theme 2: Controlled comparative synthesis of AgNPs and ZnO NPs

Gap: No study has synthesized both AgNPs and ZnO NPs from the same *Z. jujuba* extract batch under matched conditions, preventing direct mechanistic and performance comparison (Sects. 1, 8) [102].

Recommendation: A single extract batch, characterized for TPC/TFC, should be split and used for parallel synthesis of AgNPs and ZnO NPs under conditions matched as far as the differing chemistries allow.

Suggested approach: Design of Experiments (DoE) methodology, such as a Box-Behnken or Central Composite Design, applied in parallel to both systems, with shared factors (pH, temperature, extract ratio) varied identically.

11.3 Theme 3: Calcination temperature optimization for ZnO NPs

Gap: The 25–90 nm size range reported for *Z. jujuba*-ZnO NPs across three studies [24, 25, 27] has not been attributed to calcination temperature, synthesis route, or extract variability through a controlled study (Sect. 4.5, 5.1).

Recommendation: A calcination-temperature series (e.g., 300, 400, 500, 600 °C) should be performed on Zn(OH)₂ precursor from a single *Z. jujuba* extract batch, preceded by TGA/DSC analysis.

Suggested approach: XRD (Scherrer analysis with stated shape constant), BET, Tauc-method band gap (with stated transition model), and one antimicrobial and one photocatalytic assay should be performed at each calcination temperature.

11.4 Theme 4: Biomedical and ecotoxicological safety profiling

Gap: Cytotoxicity data against human cell lines exist for *Z. jujuba*-ZnO NPs (HCT-116, HepG2 [25]) but not for *Z. jujuba*-AgNPs; MIC and MBC data are absent for both systems; ecotoxicological data are absent for both systems (Sect. 7.1, 7.4, 10).

Recommendation: *Z. jujuba*-AgNPs should be evaluated against HCT-116 and HepG2 cell lines (same as [25]) using CCK-8 assay, and MIC/MBC values should be determined against a standard panel of MDR pathogens.

Suggested approach: A brine shrimp (*Artemia salina*) acute toxicity assay could be incorporated into the same study as a first-tier ecotoxicity screen without requiring specialized aquatic facilities.

11.5 Theme 5: Application-specific performance metrics

Gap: Biosensing applications of *Z. jujuba*-ZnO NPs have been demonstrated [85] without reported LOD, sensitivity, selectivity, or response time (Sect. 7.5); antioxidant assay data are absent for both *Z. jujuba*-AgNPs and *Z. jujuba*-ZnO NPs (Sect. 7.3).

Recommendation: Where biosensing or antioxidant claims are made for *Z. jujuba*-derived nanoparticles, standard performance metrics should be reported for biosensing: LOD, linear range, sensitivity, selectivity, and response time; for antioxidant: DPPH and ABTS IC₅₀ values compared against a positive control and against the bare extract.

12 Conclusion

Ziziphus jujuba Mill. leaf extract contains flavonoids, phenolic acids, triterpenoids, polysaccharides, and vitamins that have been proposed to support green synthesis of AgNPs via phyto-reduction and ZnO NPs via bio-mediated co-precipitation and calcination. The literature reviewed here, comprising approximately twelve

primary studies specific to *Z. jujuba* (Sect. 1.1), reports *Z. jujuba*-AgNPs as spherical, face-centered cubic particles of 8–30 nm with SPR activity at 420–450 nm, and *Z. jujuba*-ZnO NPs as hexagonal wurtzite particles ranging from 25 to 90 nm depending on synthesis route, with antimicrobial, photocatalytic, and in one case anticancer activity reported.

This review has identified that the existing evidence base, while informative, has not yet established (i) a controlled comparison of AgNP and ZnO NP synthesis from the same *Z. jujuba* extract batch; (ii) the cause of the substantial size range reported for *Z. jujuba*-ZnO NPs; (iii) MIC or MBC data for either nanoparticle system; (iv) cytotoxicity data for *Z. jujuba*-AgNPs comparable to that now available for the ZnO NP system; or (v) ecotoxicological or scale-up data for either system. The five thematic recommendations in Sect. 11 provide a structured basis for addressing these points in future work.

Z. jujuba represents a phytochemically well-characterized but comparatively understudied extract system for nanoparticle synthesis. The existing literature base, though limited in size, provides a workable foundation for the more systematic studies recommended here, which if implemented would substantially strengthen the evidence base and facilitate meaningful comparison of *Z. jujuba*-derived nanoparticles with those from more extensively studied plant systems.

Abbreviations

AgNPs	Silver nanoparticles
ZnO NPs	Zinc oxide nanoparticles
SPR	Surface plasmon resonance
XRD	X-ray diffraction
FTIR	Fourier-transform infrared spectroscopy
SEM	Scanning electron microscopy
TEM	Transmission electron microscopy
HRTEM	High-resolution transmission electron microscopy
SAED	Selected area electron diffraction
EDX / EDAX	Energy-dispersive X-ray spectroscopy
DLS	Dynamic light scattering
PDI	Polydispersity index
BET	Brunauer-Emmett-Teller (surface area analysis)
TGA	Thermogravimetric analysis
DSC	Differential scanning calorimetry
PL	Photoluminescence
NBE	Near-band-edge (emission)
ROS	Reactive oxygen species
MDR	Multidrug-resistant
AMR	Antimicrobial resistance
ZOI	Zone of inhibition
MIC	Minimum inhibitory concentration
MBC	Minimum bactericidal concentration
IC ₅₀	Half-maximal inhibitory concentration
DPPH	2,2-diphenyl-1-picrylhydrazyl (radical scavenging assay)
ABTS	2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (radical scavenging assay)
FRAP	Ferric reducing antioxidant power
CCK-8	Cell counting kit-8 (cytotoxicity assay)
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (cytotoxicity assay)
LOD	Limit of detection
UHPLC	Ultra-high-performance liquid chromatography
TPC / TFC	Total phenolic content / total flavonoid content

MO / MB	Methyl orange / methylene blue
JCPDS	Joint Committee on Powder Diffraction Standards
DoE	Design of Experiments
SERS	Surface-enhanced Raman scattering

Supplementary Information

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Supplementary Material 1

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Author contributions

Hamza Mustapha Umar: Conceptualization, methodology, data collection and Curation, writing the original draft preparation, literature search and synthesis. Tasiu Zangina: Supervision, writing, reviewing and editing, validation. Jamila Lamido Sumaila: Investigation, resources, writing, reviewing and editing. Akhilesh Dwivedi: Supervision, writing, reviewing and editing, formal analysis, Mohammad Aamir: review and editing, resources, validation, supervision.

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Data availability

No datasets were generated or analysed during the current study.

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Ethics and consent to participate declaration

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